AI-generated · Unofficial practice

Physics Mock

Paper 2 (Elective Part)

60 minutes（1 hour）

FOR REVISION
For revision and self-assessment only. Not an official examination paper.

INSTRUCTIONS

1. This paper consists of FOUR sections: Elective A Astronomy and Space Science, B Atomic World, C Energy and Use of Energy, D Medical Physics.
2. Attempt ALL questions in any TWO sections.
3. Each section has 8 multiple-choice questions and 1 structured question (10 marks).
4. Show all working in the structured questions. Unless otherwise specified, give numerical answers to 3 significant figures.
5. A list of data, formulae and relationships is provided on page 2.

Independent, unofficial mock paper. Not affiliated with any official examination authority.

Data, Formulae and Relationships

Data

acceleration due to gravity $g = 9.81\ \mathrm{m\,s^{-2}}$ (close to the Earth)	universal gravitational constant $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$
speed of light in vacuum $c = 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$	charge of electron $e = 1.60 \times 10^{-19}\ \mathrm{C}$
electron rest mass $m_e = 9.11 \times 10^{-31}\ \mathrm{kg}$	permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2\,N^{-1}\,m^{-2}}$
permeability of free space $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$	atomic mass unit $u = 1.661 \times 10^{-27}\ \mathrm{kg}$ （$1\,u \approx 931\ \mathrm{MeV}$）
Avogadro constant $N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$	universal gas constant $R = 8.31\ \mathrm{J\,mol^{-1}\,K^{-1}}$
Boltzmann constant $k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$	Planck constant $h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$
Stefan constant $\sigma = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$	astronomical unit $1\ \mathrm{AU} = 1.50 \times 10^{11}\ \mathrm{m}$
light year $1\ \mathrm{ly} = 9.46 \times 10^{15}\ \mathrm{m}$	parsec $1\ \mathrm{pc} = 3.09 \times 10^{16}\ \mathrm{m}$

Motion

$$v = u + at \qquad s = ut + \tfrac{1}{2}at^2 \qquad v^2 = u^2 + 2as \qquad s = \tfrac{1}{2}(u+v)t$$

Force, energy, momentum

$$F = ma \qquad F\Delta t = \Delta(mv) \qquad W = Fs\cos\theta \qquad E_P = mgh \qquad E_K = \tfrac{1}{2}mv^2 \qquad P = \frac{W}{t} = Fv$$

Circular motion and gravitation

$$a = \frac{v^2}{r} = \omega^2 r \qquad F = \frac{GMm}{r^2} \qquad v = \omega r$$

Heat and gases

$$Q = mc\Delta T \qquad Q = ml \qquad pV = nRT \qquad pV = \tfrac{1}{3}Nm\overline{c^2} \qquad E_K = \frac{3RT}{2N_A}$$

Waves

$$v = f\lambda \qquad n = \frac{\sin\theta_1}{\sin\theta_2} \qquad \sin c = \frac{1}{n} \qquad \Delta y = \frac{\lambda D}{a} \qquad d\sin\theta = n\lambda \qquad \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Electricity and magnetism

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} \qquad E = \frac{V}{d} \qquad V = IR \qquad R = \frac{\rho l}{A} \qquad P = IV = I^2R \qquad F = BIl\sin\theta \qquad F = BQv\sin\theta$$

$$B = \frac{\mu_0 I}{2\pi r} \qquad B = \frac{\mu_0 NI}{l} \qquad \varepsilon = N\frac{\Delta\Phi}{\Delta t} \qquad \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Radioactivity and nuclear energy

$$N = N_0 e^{-kt} \qquad A = kN \qquad t_{1/2} = \frac{\ln 2}{k} \qquad \Delta E = \Delta m c^2$$

Elective part

$$E = hf \qquad \tfrac{1}{2}m_e v_{\max}^2 = hf - \phi \qquad E_n = -\frac{13.6}{n^2}\ \mathrm{eV} \qquad \lambda = \frac{h}{p} \qquad \lambda_{\max} T = 2.90 \times 10^{-3}\ \mathrm{m\,K} \qquad L = 4\pi R^2 \sigma T^4$$

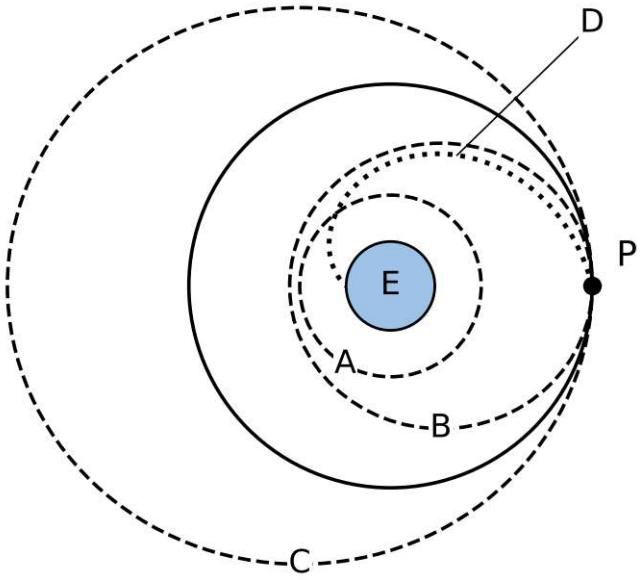
$$v = H_0 d \qquad \frac{\Delta\lambda}{\lambda_0} = \frac{v}{c} \qquad Z = \rho c \qquad I = I_0 e^{-\mu x} \qquad \alpha = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2 \qquad P = \frac{1}{f}$$

∗ Original compilation of commonly used quantities; symbols follow usual physics conventions.

Section A Astronomy and Space Science

Multiple-choice Questions

1.1. A satellite moves in a circular orbit around the Earth. On reaching point P it briefly fires its engine forwards, slightly reducing its speed, and then shuts the engine off (see the figure). Which path in the figure is the satellite most likely to follow afterwards?



A. Path A

B. Path B

C. Path C

D. Path D

1.2. Moons X and Y move in circular orbits around the same planet. X has orbital radius r and period **2.0** days; Y has orbital radius $2.25r$. What is the period of Y?

A. **3.0** days

B. **4.5** days

C. **6.75** days

D. **10.1** days

1.3. Which of the following statements about the development of the geocentric and heliocentric models is/are correct?

(1) Ancient astronomers could not observe stellar parallax, and this was used as an argument supporting the geocentric model.

(2) Based on observational data, Kepler stated that planetary orbits are ellipses with the Sun at one focus.

(3) The heliocentric model of Copernicus assumed circular planetary orbits and therefore predicted planetary positions far more accurately than the geocentric model of Ptolemy.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (1) and (2) only

1.4. The table lists the apparent magnitude m and absolute magnitude M of two stars.

Star X: $m = 2.0$, $M = -3.0$

Star Y: $m = 4.0$, $M = 1.0$

Which of the following is correct?

A. X is more luminous than Y, and X is farther from the Earth.

B. X appears dimmer than Y from the Earth.

C. X is closer to the Earth than Y.

D. Y is more luminous than X.

1.5. The parallaxes of stars P and Q are $0.050''$ and $0.020''$ respectively. The apparent brightness of Q measured on the Earth is $\frac{1}{5}$ that of P. What is the ratio $\frac{L_Q}{L_P}$ of the luminosity of Q to that of P?

A. 0.032 **B.** 0.50

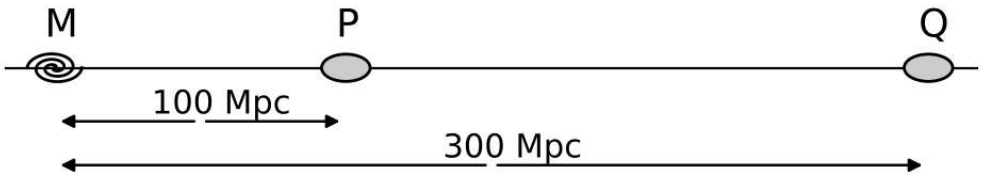
C. 1.25 **D.** 6.25

1.6. Galaxies M (the Milky Way), P and Q lie on a straight line, with P and Q at distances of **100 Mpc** and **300 Mpc** from M respectively (see the figure). The Hubble constant is $70\ \text{km s}^{-1}\ \text{Mpc}^{-1}$. Assume the galaxies move only because of the expansion of the Universe. Which of the following is/are correct?

(1) Observed from P, the recession speed of Q is $1.4 \times 10^4\ \text{km s}^{-1}$.

(2) Observed from Q, both M and P are receding.

(3) Observed from P, M is approaching P.



A. (1) only **B.** (1) and (2) only

C. (2) and (3) only **D.** (1), (2) and (3)

1.7. Which of the following information about a star can be obtained by analysing its spectrum?

(1) the chemical elements in the outer layers of the star

(2) the surface temperature of the star

(3) the distance of the star from the Earth

A. (1) and (2) only **B.** (1) and (3) only

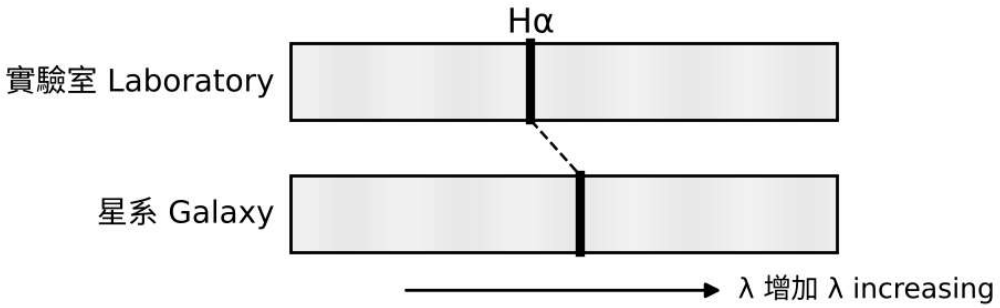
C. (2) and (3) only **D.** (1), (2) and (3)

1.8. The figure compares the hydrogen Hα absorption line in the spectrum of a laboratory source with that in the spectrum of a distant galaxy. Hα has a laboratory wavelength of $\mathbf{656.3\ nm}$ and is measured at $\mathbf{662.9\ nm}$ in the galaxy's spectrum. The Hubble constant is $\mathbf{70\ km\ s^{-1}\ Mpc^{-1}}$. Which of the following is/are correct?

(1) The galaxy is moving away from the Earth at about $\mathbf{3.0 \times 10^{6}\ m\ s^{-1}}$.

(2) The distance of the galaxy from the Earth is about $\mathbf{43\ Mpc}$.

(3) The Hβ line (laboratory wavelength $\mathbf{486.1\ nm}$) in the galaxy's spectrum is also shifted towards longer wavelength by $\mathbf{6.6\ nm}$.



A. (1) only

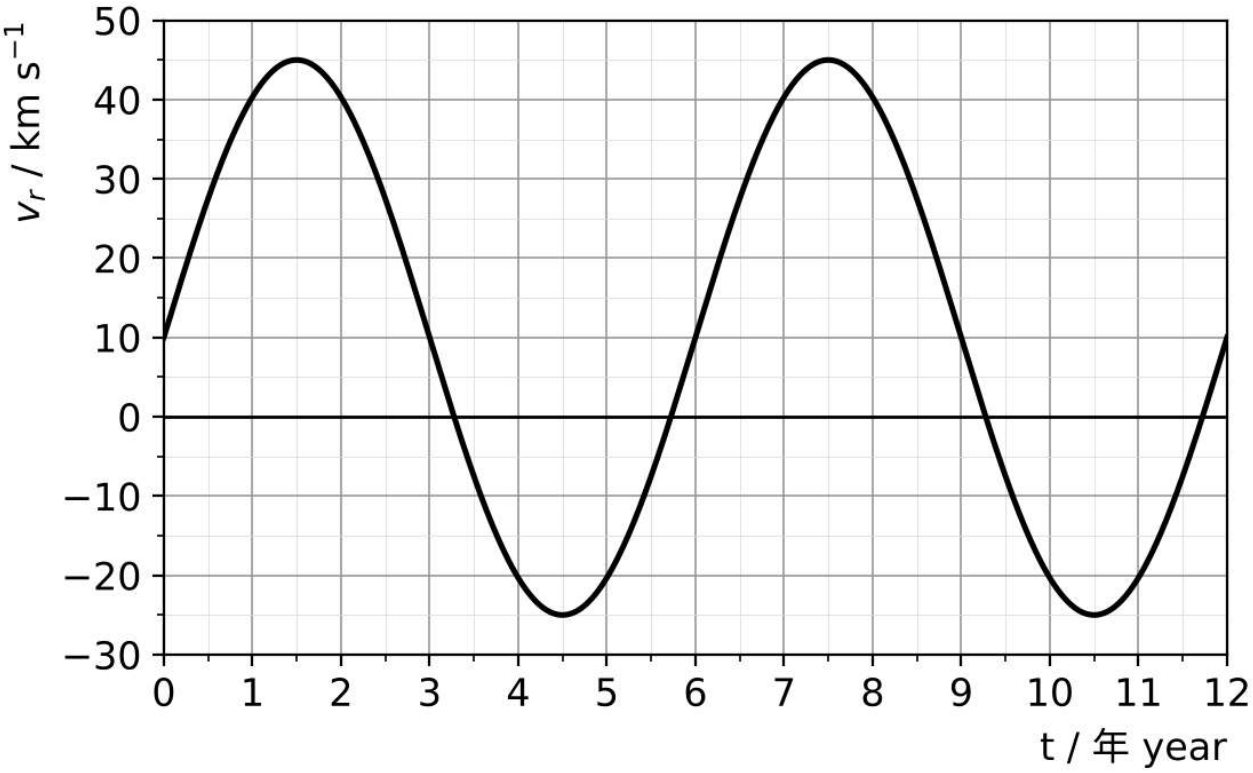
B. (2) only

C. (3) only

D. (1) and (2) only

Structured Question

1. Astronomers discover that, at a distance of $\mathbf{1500\ pc}$ from the Earth, a bright star S orbits an unseen compact object X. Assume the orbit of S is a circle centred on X, and that the mass of X is much greater than that of S. Images taken over many years show that, as seen from the Earth, the greatest angular separation between S and X is $\mathbf{0.0060''}$. Figure 2.1 shows the variation of the radial velocity $\boldsymbol{v_r}$ of S with time $\boldsymbol{t}$ (positive $\boldsymbol{v_r}$ means moving away from the Earth). Mass of the Sun $= \mathbf{1.99 \times 10^{30}\ kg}$.



(a)(i) Show that the orbital radius of S is **9.0 AU**. (1)

(ii) From Figure 2.1, read the orbital period of S, and find the mass of X in solar masses. (2)

(b)(i) From Figure 2.1, find (1) the velocity of the whole system relative to the Earth (state whether approaching or receding), and (2) the maximum change in the radial velocity of S due to its orbital motion. (2)

(ii) From the results of (a), find the orbital speed of S, and explain why it differs from the value in (b)(i)(2). (2)

(c) The maximum mass of a neutron star is about 3 solar masses. Using the result of (a)(ii), explain why astronomers believe that X is a black hole rather than an ordinary star or a neutron star. (3)

Section B Atomic World

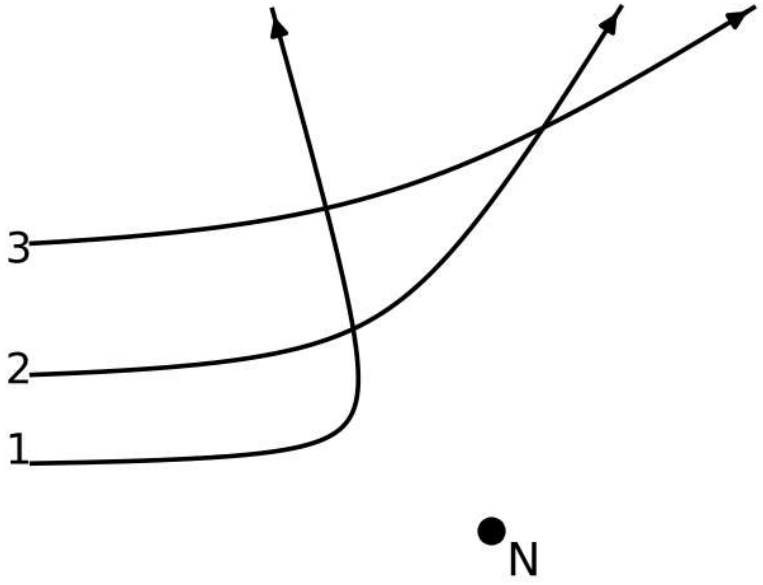
Multiple-choice Questions

2.1. Three α particles with the same kinetic energy approach a gold nucleus N along paths 1, 2 and 3 and are deflected (see the figure). Which of the following is/are correct?

(1) The α particle on path 1 comes closest to the nucleus.

(2) On each path, the kinetic energy of the α particle is least at its closest approach to the nucleus.

(3) The speed of an α particle after it has moved far away from the nucleus is smaller than before.



A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

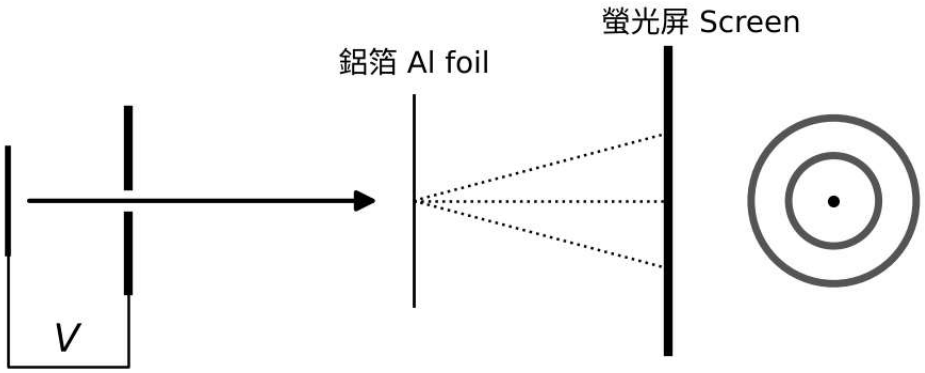
D. (1), (2) and (3)

2.2. In the set-up shown, electrons accelerated through a potential difference V pass through a very thin polycrystalline aluminium foil and form a set of concentric rings on a fluorescent screen. Which of the following is/are correct?

(1) Rings are formed because the foil consists of a large number of tiny crystals in random orientations.

(2) If V is increased from $\mathbf{4.0\ kV}$ to $\mathbf{16\ kV}$, the radii of the rings are roughly halved.

(3) If neutrons with the same speed are passed through the foil instead, the rings formed will have larger radii.



A. (1) only

B. (2) only

C. (1) and (3) only

D. (1) and (2) only

2.3. When monochromatic light of frequency f falls on a metal surface, the stopping potential of the photoelectrons is $\mathbf{1.0\ V}$. With monochromatic light of frequency $\mathbf{1.5}f$, the stopping potential is $\mathbf{2.5\ V}$. What is the work function of the metal?

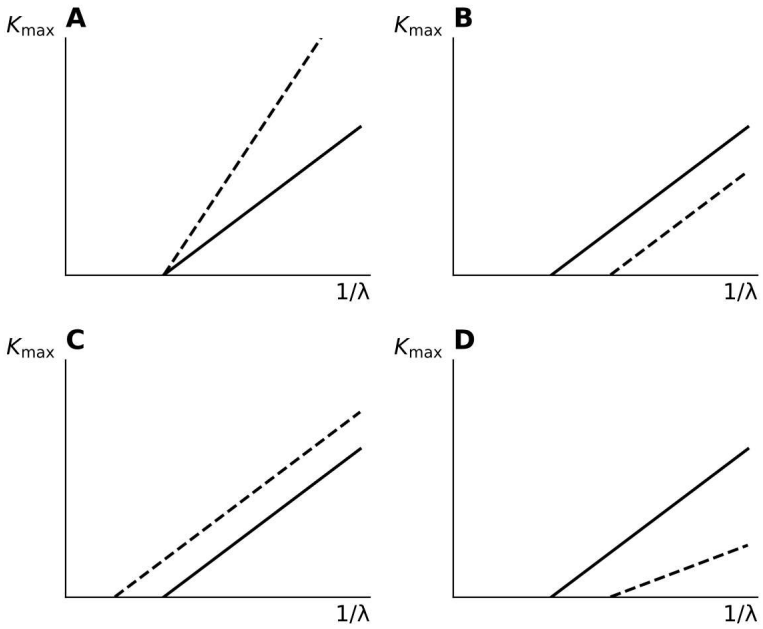
A. **1.0 eV**

B. **1.5 eV**

C. **2.0 eV**

D. **3.0 eV**

2.4. In each graph in the figure, the solid line shows the variation of the maximum kinetic energy K_{max} of photoelectrons from metal X with the reciprocal of the wavelength $\frac{1}{\lambda}$ of the incident light. Metal Y has a larger work function than X. In which graph could the dashed line represent metal Y?



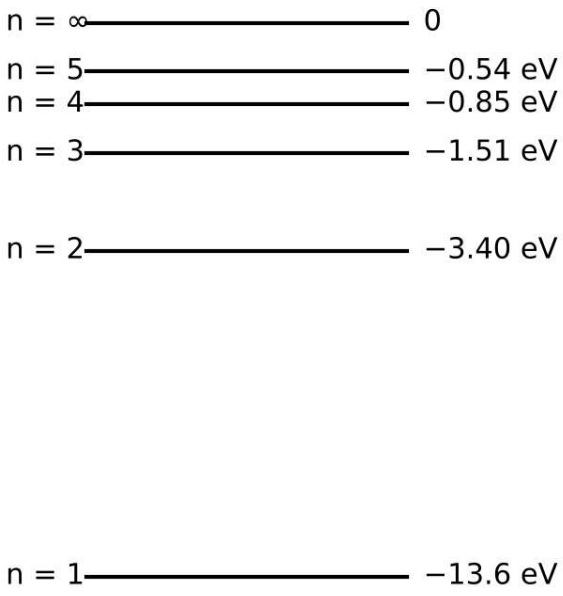
A. Graph A

B. Graph B

C. Graph C

D. Graph D

2.5. The figure shows some energy levels of the hydrogen atom. Which of the following transitions emits a photon of wavelength $1.88\ \mu m$?



A. $n = 2 \rightarrow n = 1$

B. $n = 3 \rightarrow n = 2$

C. $n = 5 \rightarrow n = 3$

D. $n = 4 \rightarrow n = 3$

2.6. An electron accelerated from rest through a potential difference of $150\ \text{V}$ has a de Broglie wavelength λ. Through what potential difference must a proton be accelerated from rest to have the same de Broglie wavelength? (The mass of a proton is about 1836 times that of an electron.)

A. $0.0817\ \text{V}$ **B.** $3.50\ \text{V}$

C. $150\ \text{V}$ **D.** $2.75 \times 10^5\ \text{V}$

2.7. Which of the following statements about the transmission electron microscope (TEM) is/are correct?

(1) Electrons accelerated through $100\ \text{kV}$ have a de Broglie wavelength of about $3.9 \times 10^{-12}\ \text{m}$ (neglecting relativistic effects), far shorter than that of visible light, so a TEM has a much higher resolution than an optical microscope.

(2) The specimen viewed in a TEM must be very thin.

(3) A TEM can be used to observe living cells immersed in water.

A. (1) only **B.** (2) only

C. (1) and (2) only **D.** (1), (2) and (3)

2.8. Which of the following statements about nanomaterials is/are correct?

(1) A solution of gold nanoparticles can appear red or purple rather than golden.

(2) If a cube of side $1\ \mu\text{m}$ is cut into small cubes of side $10\ \text{nm}$, the total surface area becomes 100 times as large.

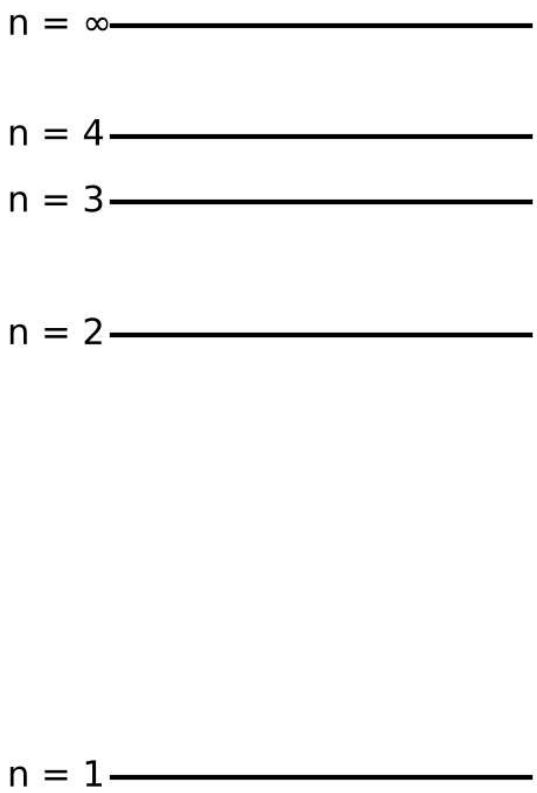
(3) A nanomaterial must have the same physical properties as the bulk material of the same chemical composition.

A. (1) only **B.** (1) and (2) only

C. (2) and (3) only **D.** (1), (2) and (3)

Structured Question

2. The energy levels of the hydrogen atom are given by $E_n = -\dfrac{13.6}{n^2}\ \text{eV}$, where $n = 1, 2, 3, \ldots$. Figure 2.2 shows the lowest few energy levels of the hydrogen atom.



(a) Explain why the spectrum of light from a hydrogen discharge tube is a line spectrum. (1)

(b) Some hydrogen atoms are excited to the $n = 3$ level.

(i) What is the greatest number of different wavelengths of photons these atoms can emit on returning to the ground state? (1)

(ii) On Figure 2.2, draw an arrow for the transition that emits the photon of the longest wavelength, and explain your choice. (2)

(iii) Find the longest wavelength in (b)(ii), and state the region of the electromagnetic spectrum to which this photon belongs. (3)

(iv) One of the atoms in the $n = 3$ level absorbs a photon and is ionized, releasing an electron with kinetic energy $0.50\ \text{eV}$. Find the wavelength of the photon, and determine whether photons of the same wavelength can ionize hydrogen atoms in the ground state. (3)

Section C Energy and Use of Energy

Multiple-choice Questions

3.1. A $12\ \text{W}$ lamp has a luminous efficacy of $90\ \text{lm W}^{-1}$. For the visible light it emits, $1\ \text{W}$ of light power corresponds to a luminous flux of $500\ \text{lm}$. What is the efficiency of the lamp in converting electrical energy into light energy?

A. 6.6% **B.** 9.0%

C. 13.2% **D.** 18.0%

3.2. Which of the following statements about wind turbines is/are correct?

(1) If the wind speed is doubled, the available wind power becomes 8 times as large.

(2) If the blade length is doubled, the available wind power becomes 4 times as large.

(3) The available wind power is proportional to the square of the air density.

A. (1) only **B.** (1) and (2) only

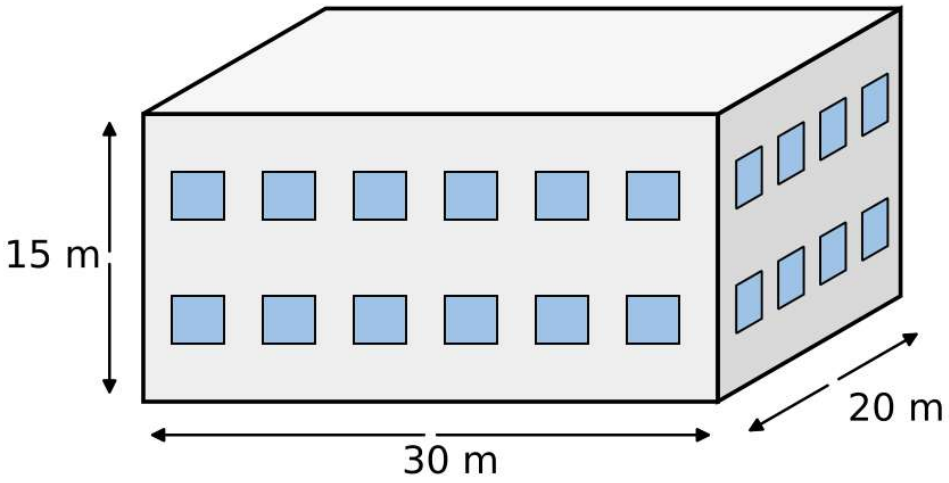
C. (2) and (3) only **D.** (1), (2) and (3)

3.3. At a certain moment, the intensity of solar radiation reaching the ground is $800\ \text{W m}^{-2}$, and the sunlight makes $30°$ with the normal to a solar panel. The panel has an area of $2.5\ \text{m}^2$ and an efficiency of 18%. What is the electrical power output of the panel?

A. $180\ \text{W}$ **B.** $312\ \text{W}$

C. $360\ \text{W}$ **D.** $1730\ \text{W}$

3.4. A rectangular office block is **30 m** long, **20 m** wide and **15 m** high (see the figure). Its four external walls have a total of **40** identical windows, each **2.5 m × 1.5 m**. The average heat transferred into the building (conduction and solar radiation) through the opaque walls and through the windows is **15 W** and **120 W** per square metre respectively. What is the overall thermal transfer value (OTTV) of the external walls of the building?



A. $15.0\ \mathrm{W\,m^{-2}}$

B. $18.2\ \mathrm{W\,m^{-2}}$

C. $25.5\ \mathrm{W\,m^{-2}}$

D. $27.0\ \mathrm{W\,m^{-2}}$

3.5. Which of the following statements about the U-value of a window is/are correct?

(1) A double-glazed window has a smaller U-value than a single-glazed window because the air layer between the panes has a very low thermal conductivity.

(2) The larger the area of a window, the larger its U-value.

(3) For the same temperature difference, the rate of heat transfer per unit area through a window is proportional to its U-value.

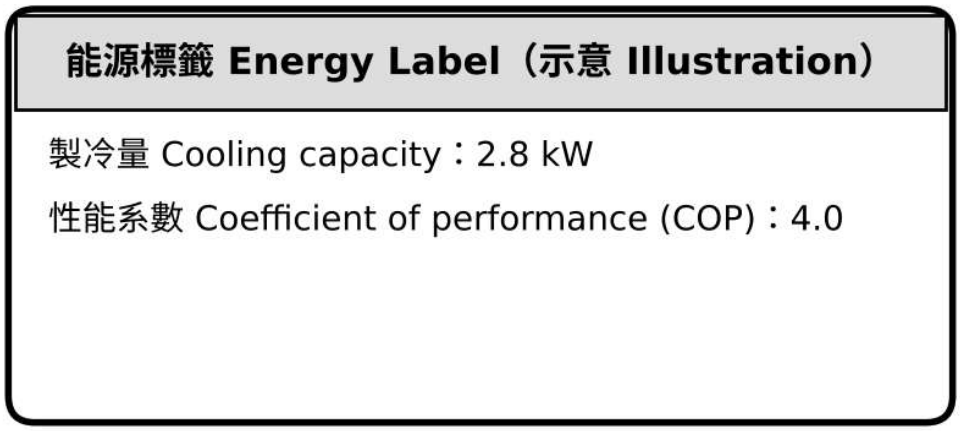
A. (1) and (3) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

3.6. The figure shows part of the information on the energy label of an air conditioner (for illustration). A family uses it for **150** days a year, **6** hours a day. Assuming the cooling capacity and coefficient of performance during operation are as shown on the label, estimate the electrical energy it consumes per year.



A. **105 kWh**

B. **158 kWh**

C. **630 kWh**

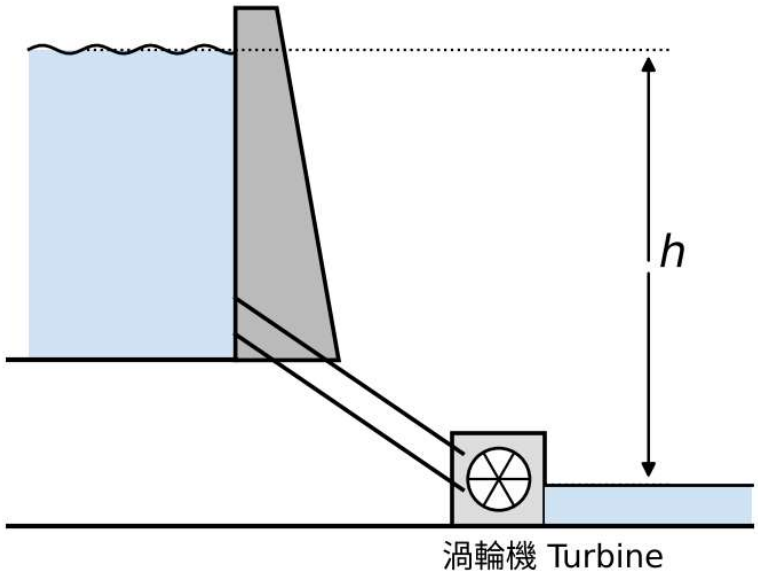
D. **2520 kWh**

3.7. The figure shows a hydroelectric power station. The height difference h between the reservoir surface and the turbine outlet is $45\ \text{m}$. Which of the following is/are correct?

(1) If all the gravitational potential energy lost by the water in falling through h became kinetic energy, the speed of the water would be about $30\ \text{m s}^{-1}$.

(2) If the volume flow rate is $20\ \text{m}^3\ \text{s}^{-1}$ and the overall efficiency is 80%, the electrical output power is about $7.1\ \text{MW}$.

(3) With h and the volume flow rate unchanged, making the pipe narrower makes the water flow faster, so the output power increases.



A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

3.8. A nuclear power station uses plutonium-239 as fuel. Its electrical power output is $900\ \text{MW}$ and its overall efficiency is 33%. The mass defect in each fission of plutonium-239 is $0.215\ \text{u}$, and the molar mass of plutonium-239 is $0.239\ \text{kg mol}^{-1}$. Estimate the mass of plutonium-239 used by the power station each day.

A. $0.318\ \text{kg}$

B. $0.964\ \text{kg}$

C. $1.46\ \text{kg}$

D. $2.92\ \text{kg}$

Structured Question

3. A transport company compares an electric bus and a diesel bus. The data are as follows:

Electric bus: battery capacity $300\ \text{kWh}$; range on a full charge $250\ \text{km}$; maximum motor output power $200\ \text{kW}$; fully loaded mass $14000\ \text{kg}$

Diesel bus: fuel consumption $35\ \text{L}$ per $100\ \text{km}$; energy content of diesel $36\ \text{MJ L}^{-1}$; fully loaded mass $13000\ \text{kg}$

(a) State one environmental advantage of using electric buses instead of diesel buses in a city. (1)

(b)(i) Find the battery energy used by the electric bus per km travelled, in MJ. (2)

(ii) Find the fuel energy used by the diesel bus per km travelled. (1)

(c) The fully loaded electric bus accelerates from rest to $\mathbf{50\ km\ h^{-1}}$ in $\mathbf{12\ s}$ on a level road.

(i) Estimate the average useful output power of the bus during this time. (Neglect resistive forces.) (2)

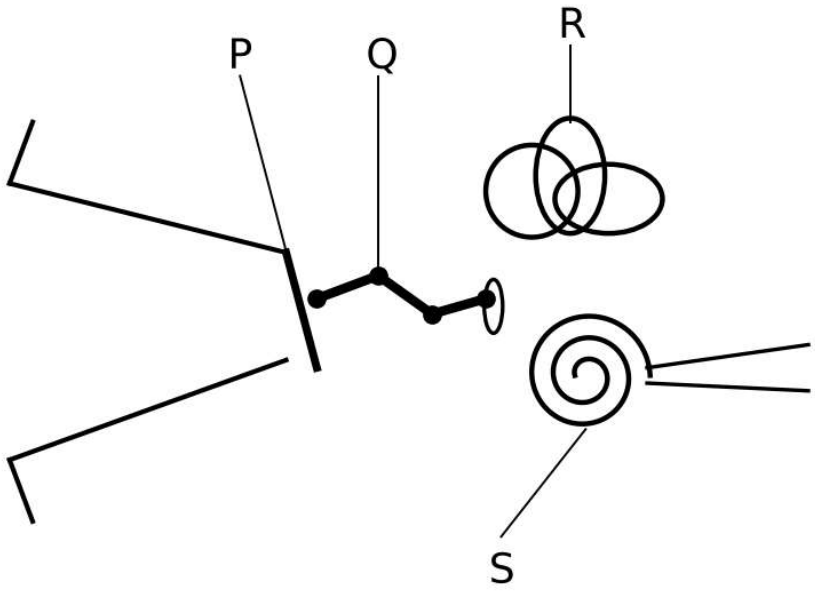
(ii) The battery-to-wheel efficiency is **85%**. Assuming the output power of the motor is constant during the acceleration, find the average power delivered by the battery during this time, and determine whether the motor can cope. (2)

(d) The electricity used to charge the battery is generated in a power station of efficiency **40%** (neglect transmission and charging losses). Someone says, "Electric buses only move fuel use from the road to the power station; they do not reduce the primary energy used." Comment on this statement using the results of (b). (2)

Section D Medical Physics

Multiple-choice Questions

4.1. The figure shows a schematic diagram of the structure of the human ear. Which part amplifies the pressure of the vibrations passed on to the inner ear?



A. P

B. Q

C. R

D. S

4.2. An optical fibre in an endoscope consists of a core and a cladding. The refractive index of the core is 1.50 and that of the cladding is 1.40. Which of the following is/are correct?

(1) The cladding prevents light from leaking from one fibre into an adjacent fibre in a bundle.

(2) The critical angle at the core–cladding boundary is about $69.0°$.

(3) The larger the angle of incidence of light entering the end face of the fibre, the more easily total internal reflection occurs at the core–cladding boundary.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

4.3. Which of the following statements about ultrasound scanning is/are correct? (The speed of ultrasound in soft tissue is $1540\ \mathrm{m\ s^{-1}}$.)

(1) If an echo is received $52\ \mu\mathrm{s}$ after the pulse is sent, the reflecting surface is about $4.0\ \mathrm{cm}$ deep.

(2) A B-scan shows the strength of an echo by the height of the echo peak.

(3) Gel is applied between the probe and the skin to reduce the reflection of ultrasound at the skin surface.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (1) and (3) only

4.4. For routine check-ups of a pregnant woman, a doctor uses ultrasound scanning rather than X-ray computed tomography (CT) to monitor the growth of the fetus. Which of the following is/are reasonable reasons?

(1) Ultrasound is not an ionizing radiation, so it is safer for the fetus.

(2) Ultrasound scanning can give real-time images showing the movement of the fetus.

(3) Ultrasound images have a higher resolution than CT images.

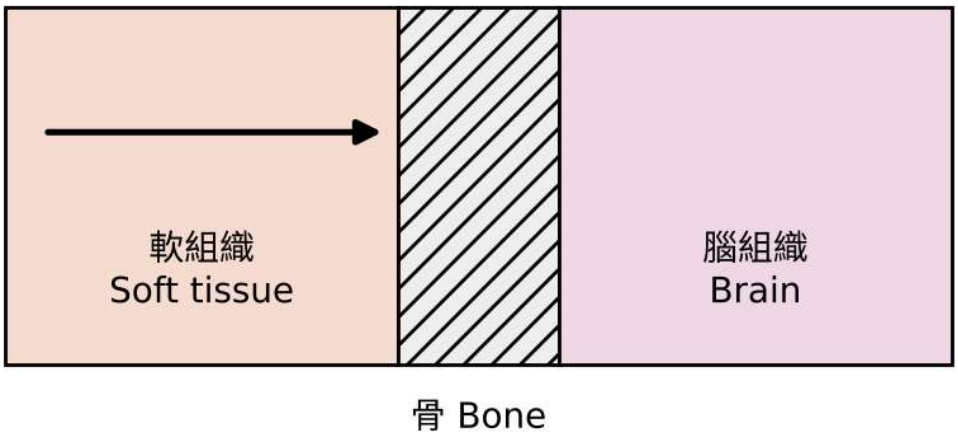
A. (1) only

B. (2) only

C. (1) and (3) only

D. (1) and (2) only

4.5. A beam of ultrasound passes normally from soft tissue into a layer of skull bone and then into brain tissue (see the figure). Soft tissue and brain tissue both have an acoustic impedance of $1.6 \times 10^6\ \mathrm{kg\ m^{-2}\ s^{-1}}$, and bone has $7.8 \times 10^6\ \mathrm{kg\ m^{-2}\ s^{-1}}$. Neglecting absorption and multiple reflections, what fraction of the incident intensity enters the brain tissue?



A. 0.130

B. 0.319

C. 0.435

D. 0.565

4.6. A beam of monochromatic X-rays has its intensity reduced to **60%** of the original after passing through an aluminium sheet **4.0 mm** thick. What additional thickness of aluminium is needed to reduce the intensity from **60%** to **15%** of the original?

A. **5.43 mm**
B. **10.0 mm**
C. **10.9 mm**
D. **14.9 mm**

4.7. Technetium-99m is a commonly used radioactive tracer. It emits only γ rays and has a half-life of **6.0** hours. Which of the following is/are correct?

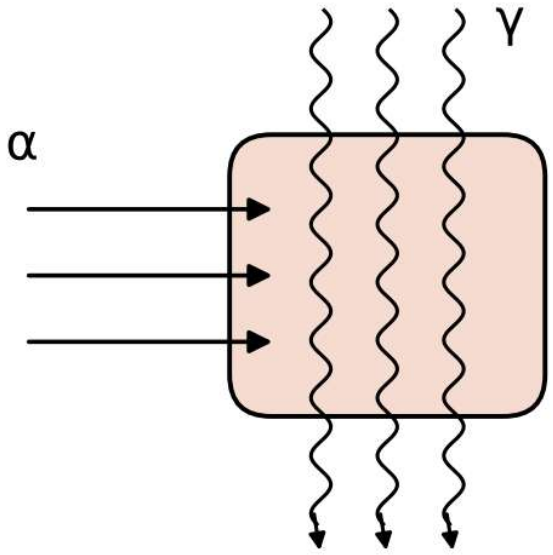
(1) **24** hours after injection, the activity of the tracer falls to **6.25%** of its initial value (ignoring any excreted from the body).

(2) The lead collimator in front of a gamma camera lets only γ rays travelling in particular directions reach the detector, so the image shows where the tracer is in the body.

(3) A tracer emitting only γ rays is used because γ rays are more strongly ionizing than α particles and so give clearer images.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

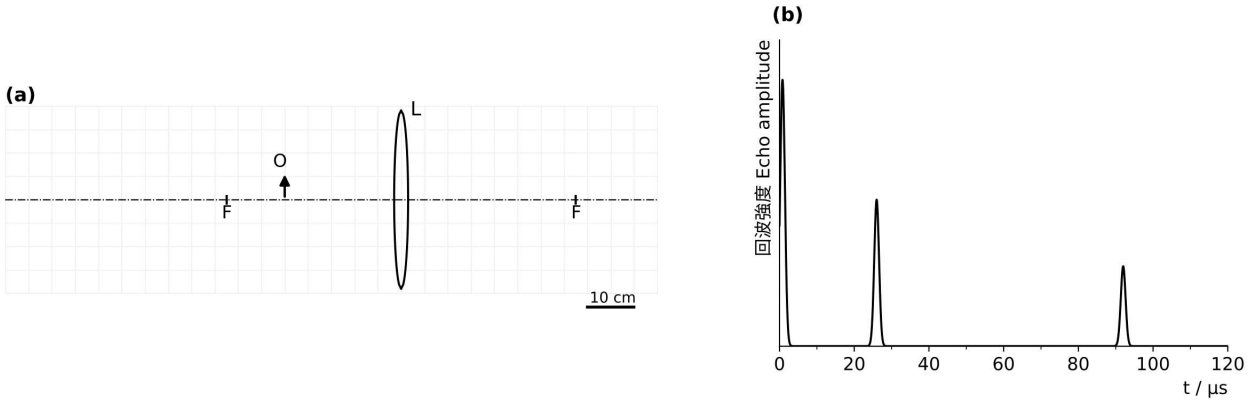
4.8. A piece of tissue of mass **0.50 kg** is irradiated by both α particles and γ rays (see the figure), absorbing **2.0 mJ** and **4.0 mJ** of energy respectively. The radiation weighting factors of α particles and γ rays are **20** and **1** respectively. What is the equivalent dose received by the tissue?



A. **12 mSv**
B. **44 mSv**
C. **88 mSv**
D. **240 mSv**

Structured Question

4. Parts (a) and (b) use diagrams (a) and (b) of Figure 2.4 respectively.



(a) A long-sighted person has her near point **75 cm** from her eyes and her far point at infinity. She wants a pair of glasses so that she can read a book **25 cm** from her eyes. (Neglect the distance between the glasses and the eyes.)

(i) Find the power of the lens required. (2)

(ii) On Figure 2.4(a), draw two rays to show how the lens forms the image of the book O, and mark the position of the image I. (2)

(iii) While wearing these glasses, what is the farthest object distance she can see clearly? Hence explain why she must look over the top of the glasses to see distant objects. (2)

(b) Figure 2.4(b) shows the display of an A-scan of the abdomen; the peak at $t = 0$ is the transmitted pulse. The speed of ultrasound in soft tissue is $1540\ \mathrm{m\ s^{-1}}$.

(i) Find the depth of the boundary that produces the echo at $t = 92\ \mu s$. (2)

(ii) Find the distance between the two boundaries that produce the two echoes. (1)

(iii) Give one reason why the echo at $t = 92\ \mu s$ is weaker than the echo at $t = 26\ \mu s$. (1)

END OF PAPER